



Date: 02/07/2025

Lab Practical #05:

Study the concept of VLAN using packet tracer.

Practical Assignment #05:

- 1. Implement the different network structures in VLAN and VLAN trunking. Also check connectivity between them using ping command or PDU utility.**

Steps to Create VLANs in Packet Tracer:

- 1. Open Packet Tracer and Create Your Topology**

- Drag a **2960 Switch** into the workspace.
- Drag PCs into the workspace and connect them to the switch using **Copper Straight-Through cables**.

- 2. Open the Switch's Configuration Window**

- Click on the switch.
- Go to the **Config** tab.

- 3. Create VLANs**

- In the **VLAN Database** section (left menu), click it.
- In the **VLAN** table:
 - Type **VLAN ID** (e.g., 10) and **Name** (e.g., Sales), then click **Add**.
 - Type **VLAN ID** (e.g., 20) and **Name** (e.g., HR), then click **Add**.

- 4. Assign Ports to VLANs**

- In the left panel, click **FastEthernet** (for example, FastEthernet0/1).
- Under **Port Mode**, select **Access**.
- In **VLAN**, select the VLAN ID you want (e.g., 10).
- Repeat for each port, assigning them to the desired VLAN.

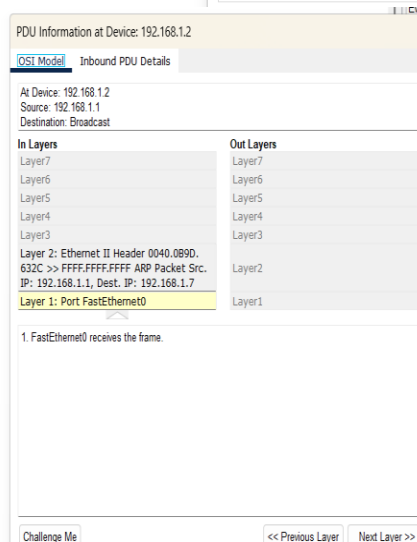
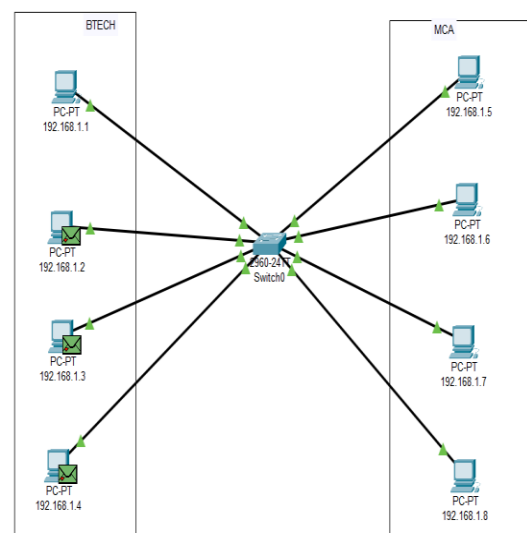
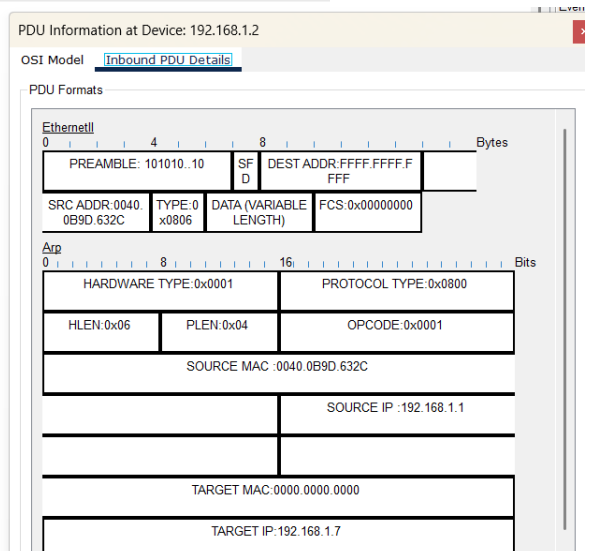
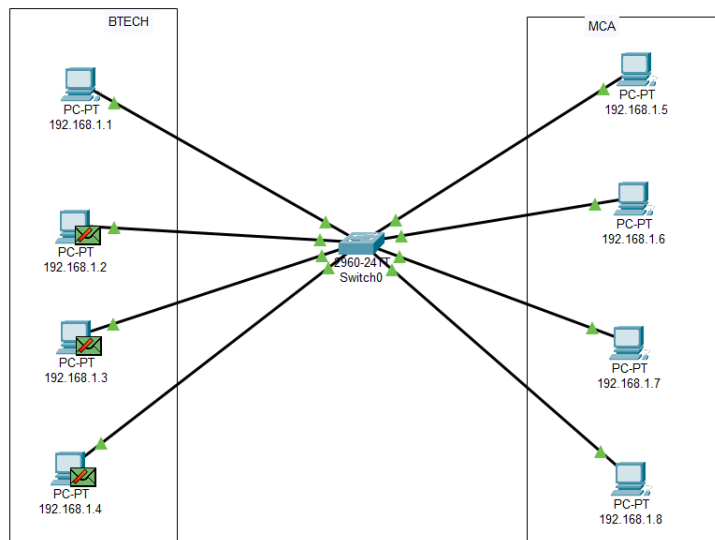
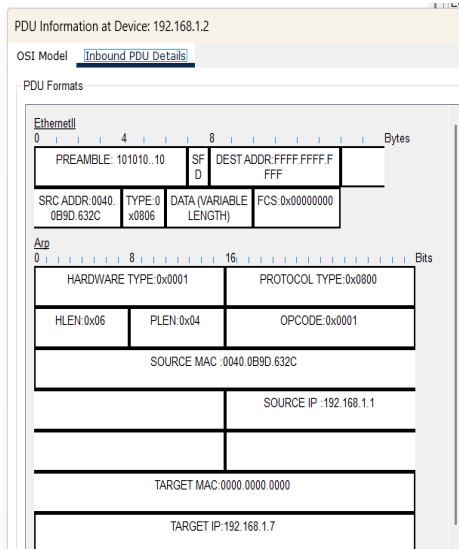
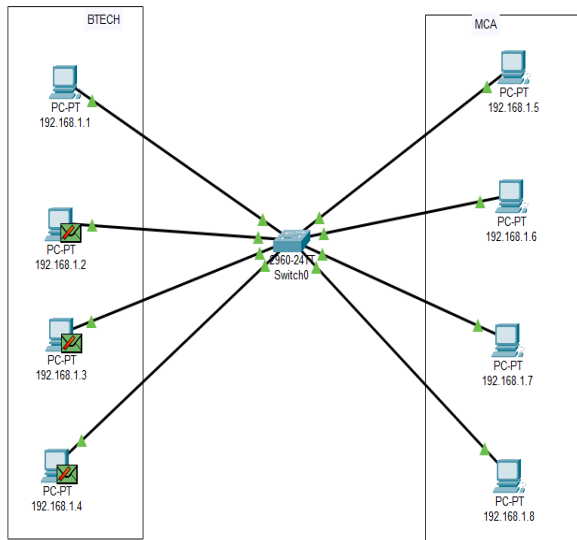
- 5. Assign IP Addresses to PCs**

- Click each PC → **Desktop** → **IP Configuration**.
- Assign IP addresses within the same network for PCs in the same VLAN.

- 6. Verify the VLAN Setup**

- PCs in the same VLAN should be able to ping each other.
- PCs in different VLANs will not communicate unless a router or Layer 3 switch is configured.

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